

Scattering Probability

Rough Calculation, for a Single Dust Particle Size

We can convert a column density of neutral H atoms to a column density of X-ray scattering dust particles, as follows. The neutral H column density in our Galaxy ¹ can be measured by measuring the total amount of 21 cm radio radiation ² that we see in a given direction through the Galaxy. All-sky maps of such radiation exist; the most recent survey is the 'HI4PI' survey (N. Ben Bekhti *et al.*, *Astron. Astrophys.*, **594**, A116 (2016)).

The average 'gas to dust' mass ratio of the interstellar medium is of order 100- meaning, the mass density of interstellar dust is about 1% of the mass density of gas. Let's call the mass density of dust ρ_{dust} , that of gas ρ_{gas} . Most of the interstellar gas is Hydrogen (and most of it is neutral when you average over large regions), so the mass of a gas particle is about one proton mass, m_p . The mass of a dust grain is $m_g = (4\pi/3)a^3\rho_{\text{grain}}$, where a is the radius of the dust particle (we're assuming they are spherical), and ρ_{grain} is the mass density of the grain material. So then, if we call N_{H} the column density of neutral H, we have for the column density of dust grains, N_{d} :

$$N_{\text{d}} = \frac{1}{100} \frac{N_{\text{H}} m_p}{m_g} \quad (1)$$

Dust particles in our Galaxy have radii between $\sim 0.1 - 0.3 \mu\text{m}$. A lot of the dust is Silicate, and a characteristic value for the grain material density is $\rho_{\text{grain}} = 3 \text{ g cm}^{-3}$. A good starting value for the gas density of the ISM is $n \sim 1 \text{ H atom cm}^{-3}$. When we multiply this column density of dust grains with the cross section for scattering of X-rays, we get the optical depth of the dust to scattering, $\tau_{\text{scattering}}$. The fraction of photons scattered by a column density of dust is $1 - \exp(-\tau_{\text{scattering}})$. So now we need the cross section for scattering of X-rays by dust grains.

¹neutral H is also referred to as 'H I' or 'H-one' by astronomers; traditional spectroscopic nomenclature has each ionization stage of each element numbered with a Roman numeral, starting with 'I' for the neutral species. So O VIII is oxygen with one electron left, or 'H-like oxygen', as the physicists usually say.

²The proton and the electron both have an intrinsic magnetic moment. When the two moments are aligned in an atom, the atom has slightly more energy than when they are anti-aligned. The transition between the two involves an amount of energy equal to that carried by a photon of radiation of wavelength 21.106 cm

The scattering cross section can be calculated from classical electrodynamics. In the simplest approximation, you can just do 'wave optics', and calculate the diffraction pattern of a dust grain. The intensity in the diffraction pattern, as a function of direction, is proportional to the probability that a photon will be scattered into that direction. That calculation, for spherical grains of radius a and density ρ_{grain} , radiation wavelength λ (or photon energy $E = hc/\lambda$) gives

$$\begin{aligned}\frac{d\sigma_{\text{scattering}}}{d\Omega} &= 2a^2 \left(\frac{2\pi a}{\lambda}\right)^4 |m-1|^2 \left(\frac{j_1(x)}{x}\right)^2 (1 + \cos^2 \theta) \\ x &= \left(\frac{4\pi a}{\lambda}\right) \sin\left(\frac{\theta}{2}\right)\end{aligned}\tag{2}$$

(Mauche and Gorenstein, *Astrophys. J.*, **302**, 371 (1986)). Here, θ is the angle through which the photon trajectory is deflected by the scattering (and so $\theta = 0$ means: no deflection). The function $j_1(x)$ is the spherical Bessel function of order 1. The quantity m is the index of refraction of the grain material, which depends on the wavelength of the radiation. You should interpret the expression in Eq.(2) as follows: $d\sigma/d\Omega$ is the *differential scattering cross section*, in units of surface area per unit solid angle (so $\text{cm}^2 \text{sr}^{-1}$; steradian (sr) is the unit of solid angle). You recall that the cross section is one of the factors that goes into the calculation that a photon will be scattered as it traverses a region of space (of length L along the photon path and density n of scattering particles, or column density nL). Here, we have the full recipe to figure out what the probability is that a photon will be scattered in a given range of directions. Choose an outgoing direction, and a small range of angles around that direction—that means, choose a small 'piece of the sky' of solid angle $d\Omega$ as seen from the grain. Then the probability that a photon will go off and be scattered into that small 'region of the sky' is proportional to $d\sigma_{\text{scattering}}/d\Omega$ for that direction, multiplied by the size (solid angle) of the patch of sky you chose around that direction.

The probability that a photon will be scattered into *any* direction is the integral of $d\sigma_{\text{scattering}}/d\Omega$ over the entire sky (over all angles). When you integrate Eq(2). over all angles, you get

$$\sigma_{\text{scattering}} = 2\pi a^2 \left(\frac{2\pi a}{\lambda}\right)^2 |m-1|^2\tag{3}$$

Let's make this a little more definite. Choose a Cartesian coordinate system, with the scattering particle at the origin. The incoming photon propagates up along the z -axis; it is scattered at the origin. The scattering angle θ is the polar angle between the direction of travel of the scattered photon and the positive z -axis. The scattering geometry is azimuthally symmetric around the z -axis, so the scattering probability does not depend on the azimuthal angle ϕ (that is the angle between the projection of the outgoing photon direction onto the $x - y$ -plane, and the positive x -axis). Now pick a range of angles $\theta \rightarrow \theta + d\theta$. Those angles all fall in a narrow annulus around the sky, of width $d\theta$. To figure the solid angle of this annulus, assume you have a sphere of radius r centered at the origin. Projected on this sphere, the radius (in a plane of constant θ or, equivalently, constant z) of the annulus we drew above is equal to $r \sin \theta$; its width is $rd\theta$. Its surface area is $2\pi \times \text{radius} \times \text{width} = 2\pi r^2 \sin \theta d\theta$. By definition, the solid angle of a surface area A seen at a distance r is equal to A/r^2 (so the solid angles of the Sun and the Moon as seen from Earth are almost exactly equal, despite the differences in A and r for both objects: they fill the same 'area on the sky'). Therefore, the solid angle of our annulus is

$$d\Omega = 2\pi \sin \theta d\theta \quad (4)$$

You can figure the solid angle of any area on the sky between two angles θ_1 and θ_2 by integrating:

$$\Omega = \int_{\theta_1}^{\theta_2} d\Omega = 2\pi \int_{\theta_1}^{\theta_2} \sin \theta d\theta \quad (5)$$

and for the entire sky ($\theta_1 = 0, \theta_2 = \pi$), you find $\Omega_{\text{full sky}} = 4\pi$ steradians.

Here we go: when a photon traverses an infinitesimal distance dl through a medium of scattering particles of density n per unit volume, the probability that the photon will be scattered into the range of angles $\theta \rightarrow \theta + d\theta$ is given by

$$d\tau = n dl \left(\frac{d\sigma_{\text{scattering}}}{d\Omega} \right) d\Omega = n dl \left(\frac{d\sigma_{\text{scattering}}}{d\Omega} \right) \cdot 2\pi \sin \theta d\theta \quad (6)$$

Check the dimensions: $d\sigma_{\text{scattering}}/d\Omega$ has the unusual dimensions 'cm² per steradian', $2\pi \sin \theta d\theta$ is solid angle in steradians, n is particles per cm³, and

dx is cm, so $d\tau$ is just a number, as it should be. From the optical depth along dx you can calculate the scattering probability from: probability = $1 - \exp(-\text{optical depth})$. If the optical depth is very small, you just get probability $\approx$ optical depth (optical depth $\ll 1$).

A few practical remarks. For most applications we are interested in, the argument x of the Bessel function in Eq.(2) is very small, and you can approximate the formula for $d\sigma/d\Omega$ by a Gaussian distribution. Mauche and Gorenstein (1986; their Eqs. (5), (6), (10), (11)) list some convenient approximations.

Finally, for low photon energies, large grain sizes, and large scattering angles, the approximations based on simple diffraction for calculating the intensity pattern of the scattered radiation break down. In that case, you have to go back to Maxwell's equations for the electromagnetic field (the incoming radiation, the field inside the scattering particle, the field radiated by the material in the particle), and you have to match these fields on the surface of the scattering particle using the proper boundary conditions on **E**, **D**, **B**, **H**. For spherical particles, you can use a mathematical trick (any function of the two angles θ and ϕ in spherical coordinates can be written as a sum over the so-called spherical harmonic functions, so then it's easy to write the boundary conditions on the surface of the particle), and you can numerically solve the resulting algebraic equations. That is referred to as the *Mie solution*. The result, compared to Eq.(2) is generally that the cross sections (scattering probabilities) end up being overestimated by a factor of a few. Details are in Lia Corrales' paper: Corrales, *Astrophys. J.*, **805**, 23 (2015), and references therein.

Practical Application

Let's adopt the approximation that $x \ll 1$ in Eq.(2). Mauche and Gorenstein calculate that the integrated scattering cross section (Eq.(3)) in that case becomes

$$\sigma_{\text{scattering}} = 6.3 \times 10^{-11} \left(\frac{2Z}{M} \right)^2 \left(\frac{\rho_{\text{grain}}}{3} \right)^2 \left(\frac{a}{0.1 \mu m} \right)^4 \left(\frac{E}{\text{keV}} \right)^{-2} \left(\frac{F(E)}{Z} \right)^2 \text{ cm}^2 \quad (7)$$

Symbols: Z is nuclear charge of the atoms in the dust material, M the

atomic mass of those atoms, ρ_{grain} the mass density of the dust grains (not the mass density of the ISM in dust grains, but the density of the grain material itself), a the radius of the grains, E the photon energy. The function $F(E)$ contains the so-called atomic scattering functions, and far away from resonances, $|F(E)| \approx Z$. Let's assume a reference photon energy of 1 keV (at lower energies, absorption also becomes important; at higher energies, the scattering probability decreases rapidly). Let's also assume the reference values for the other quantities. Then the optical depth to scattering is

$$\tau_{\text{scattering}} = N_{\text{d}} \sigma_{\text{scattering}} \quad (8)$$

with N_{d} the column density of dust particles (particles cm^{-2}). We get, using the conversion in Eq.(1):

$$\tau_{\text{scattering}} = 8.4 \times 10^{-23} N_{\text{H}} \left(\frac{a}{0.1 \mu\text{m}} \right) \left(\frac{\rho_{\text{grain}}}{3} \right) E_{\text{keV}}^{-2} \quad (9)$$

Scattering Probability

Calculation for a Realistic Distribution of Dust Particle Sizes

The scattering cross section depends strongly on the particle size, and we know that the particle sizes in the ISM have a wide distribution. We should average over that distribution when we compute the cross section, and also the conversion from dust number (particle) density to mass density.

An old, but simple and still useful expression for the size distribution is the one given by Mathis, Rumpl, and Nordsieck (*Astrophys. J.*, **217**, 425 [1977]):

$$\frac{dn_{\text{d}}}{da} = A a^{-3.5} \quad (10)$$

where n_{d} is the number density (grains per unit volume), a the grain radius, dn_{d}/da the number density of grains with radii between a and $a + da$ (grains per unit volume, per unit a). The distribution shows that there are lots of small grains, and very few large grains. We will have to choose a minimum and maximum grain size for the calculation.

First the total number density:

$$\begin{aligned}
n_d &= \int_{a_{\min}}^{a_{\max}} \frac{dn_d}{da} da = \int_{a_{\min}}^{a_{\max}} A a^{-3.5} da = \\
&= \frac{A}{2.5} \frac{1}{a_{\min}^{2.5}} \left(1 - \left[\frac{a_{\min}}{a_{\max}} \right]^{2.5} \right)
\end{aligned} \tag{11}$$

(we're anticipating that $a_{\min}/a_{\max} \ll 1$).

Now the mass density:

$$\begin{aligned}
\rho_d &= \int_{a_{\min}}^{a_{\max}} \frac{4\pi}{3} a^3 \rho_{\text{grain}} \frac{dn_d}{da} da = \int_{a_{\min}}^{a_{\max}} A \frac{4\pi}{3} a^3 \rho_{\text{grain}} a^{-3.5} da = \\
&= \frac{8\pi A}{3} \rho_{\text{grains}} a_{\max}^{0.5} \left(1 - \left[\frac{a_{\min}}{a_{\max}} \right]^{0.5} \right)
\end{aligned} \tag{12}$$

where ρ_{grain} is the mass density of the grain material (most of it is silicates, so $\rho_{\text{grain}} \approx 3 \text{ g cm}^{-3}$).

Now we can fix A : assuming $a_{\min}/a_{\max} \ll 1$, we have

$$\begin{aligned}
\rho_d &= \frac{8\pi A}{3} \rho_{\text{grains}} a_{\max}^{0.5} = \\
&= \frac{1}{G} \rho_{\text{gas}}
\end{aligned} \tag{13}$$

where ρ_{gas} is the interstellar gas density, and G is the so-called *gas-to-dust mass ratio*. The consensus is that on average $G \sim 100$. So then we get ($\rho_{\text{gas}} = n_{\text{H}} \mu m_p$, with n_{H} the number density of H atoms, m_p the mass of the proton (almost equal to the mass of a H atom), and μ the 'mean molecular weight', *i.e.* the average mass of the gas particles in units of the mass of an H atom; for the standard astrophysical mixture of elements (90% H by number, 10% He by number, tiny amounts of the heavy elements), $\mu = 1.4$)

$$A = \frac{3\mu m_p n_{\text{H}}}{8\pi \rho_{\text{grain}} a_{\max}^{0.5} G} \tag{14}$$

Now the optical depth to scattering. Particles with sizes between a and $a + da$ contribute to the total scattering optical depth:

$$d\tau_{\text{scattering}} = \frac{dn_d}{da} da \cdot \sigma_{\text{scattering}} \cdot L \quad (15)$$

over a distance L . Using Eq.(7) for the cross section (the largest particles contribute much more than the smaller ones), and integrating over a :

$$\tau_{\text{scattering}} = \frac{2}{3} \frac{A\sigma_0 E_{\text{keV}}^{-2} L}{a_0^4} a_{\text{max}}^{1.5} \left(1 - \left[\frac{a_{\text{min}}}{a_{\text{max}}} \right]^{1.5} \right) \quad (16)$$

Here, $\sigma_0 = 6.3 \times 10^{-11} \text{ cm}^2$, $a_0 = 0.1 \mu\text{m}$ in cm (from Eq.(7)). You can see what's going to happen: the expression for A contains the density of H atoms, n_H , so, with L in the numerator of the expression for $\tau_{\text{scattering}}$, we'll find that $\tau_{\text{scattering}}$ is proportional to N_H , the column density of H atoms.

Putting in the expression for A , I get

$$\tau_{\text{scattering}} = \frac{1}{G} \mu m_p N_H \frac{a_{\text{max}}}{4\pi \rho_{\text{grain}}} \frac{\sigma_0 E_{\text{keV}}^{-2}}{a_0^4} \quad (17)$$

Now we choose $G = 100$, $\rho_{\text{grain}} = 3 \text{ gr cm}^{-3}$, $\mu = 1.4$, $a_0 = 0.1 \mu\text{m} = 10^{-5} \text{ cm}$, $a_{\text{max}} = 0.3 \mu\text{m} = 3 \times 10^{-5} \text{ cm}$, and $m_p = 1.67 \times 10^{-24} \text{ gr}$:

$$\tau_{\text{scattering}} = 1.17 \times 10^{-22} N_H E_{\text{keV}}^{-2} \quad (18)$$

Photoelectric Absorption Probability

We should also look at the photoelectric absorption probability: for large N_H , the scattering probability goes up, but once N_H becomes large enough, the probability of a photon being absorbed by the atoms in the ISM exceeds the X-ray scattering probability, and we won't see X-rays. The probability for a photon to be absorbed by an atom by photoelectric absorption (which removes an electron entirely from the atom) scales, very roughly, like E^{-3} , with E the photon energy. So let's choose $E = 1 \text{ keV}$

again as a convenient reference energy. At that energy (and high ³ energies in general), the biggest contributor to the photoelectric absorption probability of the interstellar gas is in fact neon, with significant contributions from iron (out of the $n = 2$ shell), and oxygen. The cross section at 1 keV for photoelectric absorption of interstellar gas of 'standard' composition is $\sigma_{\text{phot, ISM}} = 2.4 \times 10^{-22} \text{ cm}^2$, so

$$\tau_{\text{phot, ISM}} \approx 2.4 \times 10^{-22} N_{\text{H}} E_{\text{keV}}^{-3} \quad (19)$$

³compared to the ionization energy of the lowest energy state of the Hydrogen atom, $E_{\text{H}} = 13.6 \text{ eV}$

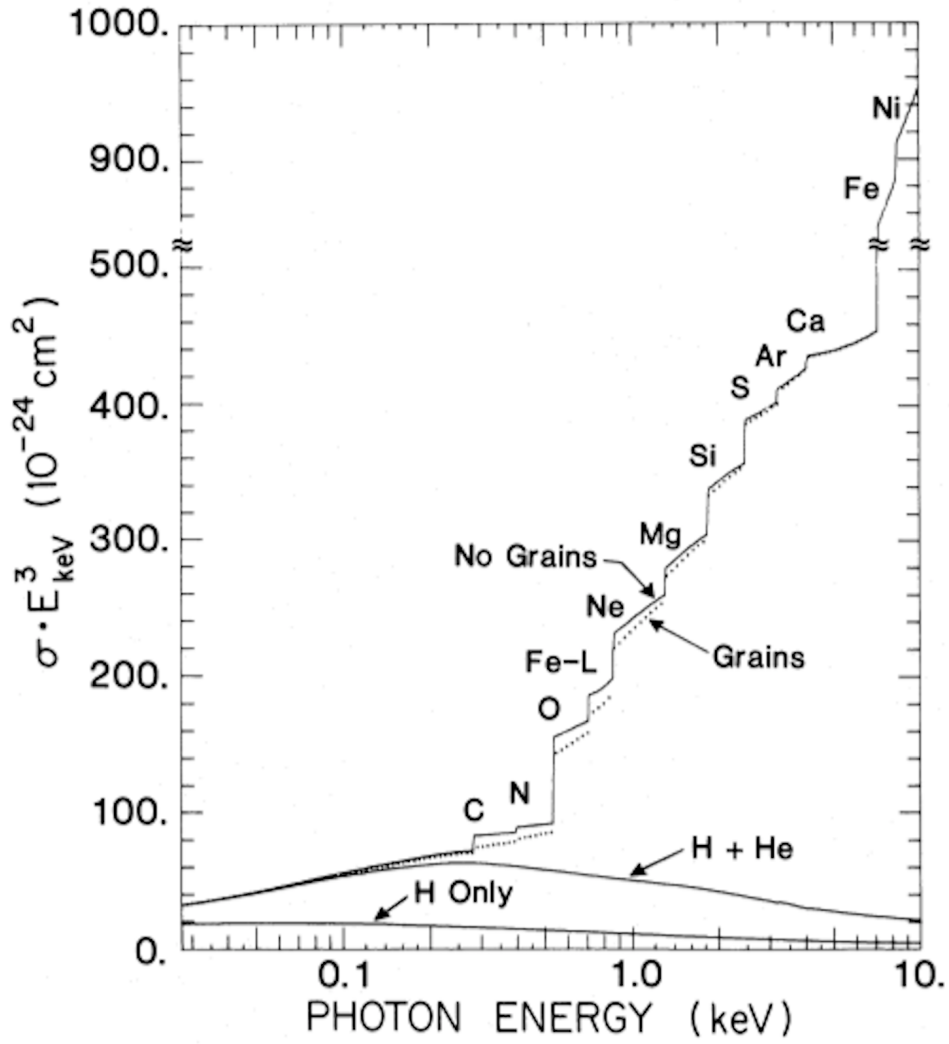


Figure 1: The photoelectric absorption cross section for interstellar gas of cosmic composition (Morrison and McCammon, *Astrophys. J.*, **270**, 119 (1983)).